Participation-Weighted Fair Value Estimation in Limit Order Books

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Abstract

The standard microprice weights bid and ask sides by order quantity. We propose weighting by order count, the number of independent orders, arguing that count captures the diversity of market consensus which quantity alone misses. Testing across 183 Indian equities and 2 index futures contracts using 5-level limit order book data, we find that the count-based microprice outperforms the quantity-based microprice by 30 to 38% in rank IC, with statistical significance of $p < 10^{-4}$ (paired t-test, $t = 18.8$ at the 10-tick horizon). The advantage holds across spread regimes, persists on each test day independently, survives an order fragmentation control, and extends to index futures at deeper book levels. We further show that signed order count flow predicts returns in the correct direction (IC $= +0.032$), while the standard signed volume flow predicts the wrong direction (IC $= -0.021$). These findings are consistent with Rosu's (2009) theoretical prediction that equilibrium prices depend on the *numbers* of orders, not their sizes. Code and anonymised data are available at `github.com/MehulBandhu/lob-microstructure`.

1. Introduction

Estimating the fair value of a traded instrument from limit order book (LOB) data is the central problem in electronic market making. The fair value determines quote placement, inventory management, and adverse selection control. A market maker whose fair value estimate is even 0.5 basis points more accurate than a competitor's can quote tighter when correct and wider when uncertain, compounding into significant P&L over millions of daily quotes.

The dominant approach is Stoikov's microprice [1], which adjusts the mid-price using the bid-ask imbalance weighted by quantity:

$$\hat{V}_{\text{qty}} = \frac{P_1^b \cdot Q_1^a + P_1^a \cdot Q_1^b}{Q_1^b + Q_1^a} \tag{1}$$

where P_1^b, P_1^a are the best bid and ask prices and Q_1^b, Q_1^a the corresponding quantities. This is widely used in practice and has a martingale justification under Markov assumptions on the book state [1].

However, quantity conflates two fundamentally different book states. Consider 10,000 shares on the bid from 2 orders versus 10,000 shares from 200 orders. The quantity-weighted microprice treats these identically. We argue they carry very different information: 200 independent participants represent robust consensus about fair value, while 2 large orders represent concentrated, and potentially fragile, positioning.

We propose a **participation-weighted microprice** that replaces quantity with order count:

$$\hat{V}_{\text{cnt}} = \frac{P_1^b \cdot N_1^a + P_1^a \cdot N_1^b}{N_1^b + N_1^a} \tag{2}$$

where N_1^b, N_1^a are the number of orders at the best bid and ask. This is grounded in Rosu's (2009) equilibrium model of the limit order book [2], which proves that "in equilibrium, the bid and ask prices depend only on the *numbers* of buy and sell orders in the book." The theoretical intuition connects to Condorcet's Jury Theorem: if each order represents an independent signal about fair value, the aggregate accuracy increases with the number of independent signals, not their total magnitude.

I built a production tick capture system on AWS Lightsail (Linux instance located in Mumbai) that collects 5-level LOB data across 3000 instruments via the Zerodha Kite WebSocket API. The system uses a double-buffered Parquet writer with background flush threads and handles reconnects, circuit breaking, and post-market consolidation. This provided snapshots of orderbook every second for every instrument. I used data collected in the first week of March. This collector is live, and will continue collecting data for months.

2. Data and Methodology

2.1 Data

We use tick-level LOB data from 183 NSE equities (97 NIFTY 100 constituents and 86 midcap stocks) and 2 index futures contracts (NIFTY and BANKNIFTY front-month), collected over 5 trading days (March 4 to 10, 2026). Each tick contains 5 levels of depth on both sides: prices, quantities, and order counts at each level. After filtering for market hours (9:15 to 15:30 IST) and valid quotes, the dataset comprises approximately 11.2 million equity ticks and 273,000 futures ticks.

2.2 Fair Value Estimators

We define seven estimators (Table 1). The multi-level variants use exponential decay weights $w_i = e^{-\lambda i} / \sum_j e^{-\lambda j}$ with $\lambda = 0.5$ across levels $i = 1, \ldots, 5$.

Estimator	Weighting	Levels	Formula
Mid	None	n/a	$(P^a + P^b)/2$
Qty-L1	Quantity	L1	Standard microprice (1)
Cnt-L1	Order count	L1	Participation-weighted (2)
Hyb-L1	Qty × Count	L1	Diversity-scaled quantity
Qty-L5	Quantity	L1 to L5	Depth-weighted microprice
Cnt-L5	Order count	L1 to L5	Multi-level participation
Hyb-L5	Qty × Count	L1 to L5	Multi-level blend

Table 1: Seven fair value estimators tested.

2.3 Evaluation Protocol

For each instrument, the first 3 days serve as warmup and the last 2 days as test. Evaluation is strictly causal: each FV is computed at tick t using only the book state at t, and compared to the realised mid-price at $t + H$ for $H \in \{10, 20, 50, 100\}$ ticks (approximately 10s to 100s). Samples are non-overlapping (every H ticks) to prevent autocorrelation inflation.

We report two metrics. **MSE Ratio:** $\text{MSE}(\hat{V}, M_{t+H})/\text{MSE}(M_t, M_{t+H})$. A ratio below 1.0 means the FV is a better forecast of the future mid than the current mid. **Rank IC:** Spearman correlation between the FV displacement $(\hat{V}_t - M_t)$ and the forward return $(M_{t+H} - M_t)$, measuring whether the FV correctly predicts the direction and relative magnitude of the next price move.

3. Results

3.1 Fair Value Comparison

Table 2 presents the main results across all 185 instruments. Figure 1 shows the IC curves and per-stock distribution.

	$H = 10$		$H = 20$		$H = 50$		$H = 100$	
	MSE	IC	MSE	IC	MSE	IC	MSE	IC
Qty-L1	1.022	+.112	1.010	+.077	1.010	+.047	1.004	+.042
Cnt-L1	**1.001**	**+.129**	**1.000**	**+.096**	1.001	**+.062**	1.001	+.045
Hyb-L1	1.031	+.122	1.015	+.086	1.012	+.052	1.006	+.041
Qty-L5	1.076	+.144	1.021	+.112	1.013	+.072	1.009	+.056
Cnt-L5	1.352	**+.189**	**0.996**	**+.149**	**0.999**	**+.099**	**1.000**	**+.074**
Hyb-L5	1.168	+.146	1.056	+.116	1.027	+.079	1.017	+.060

Table 2: Fair value estimator comparison across 185 instruments. Bold indicates best in class. Cnt-L5 achieves MSE < 1.0 at $H \geq 20$ and the highest IC at every horizon.

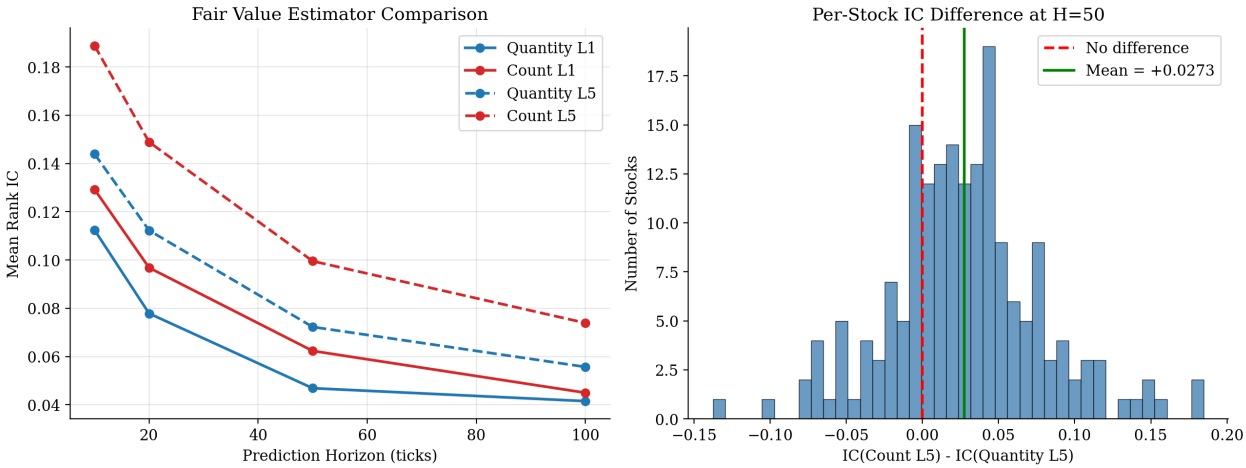

Figure 1: Count-based fair value consistently outperforms quantity-based across all horizons (left). The per-stock IC difference is positive for 74% of instruments at H=50 (right).

Cnt-L5 achieves the highest IC at every horizon: +0.189 at $H = 10$, +0.099 at $H = 50$, +0.074 at $H = 100$. It is the only multi-level estimator with MSE below 1.0 at $H \geq 20$, meaning it forecasts the future mid-price more accurately than the current mid-price itself. The improvement over quantity-based is 31% at $H = 10$, 37% at $H = 50$, and 32% at $H = 100$.

The order count finding was unexpected. I initially included bid_orders as one of 17 features in a Ridge regression for return prediction and noticed its coefficient was larger than the quantity-based OBI. This led me to test the count-based microprice directly, which turned out to be the strongest result of the project.

3.2 Statistical Significance

At L5, the count advantage is statistically significant at every horizon ($p < 0.004$). At L1, significance holds through $H = 50$ ($p = 0.0006$) but vanishes at $H = 100$ ($p = 0.58$), consistent with L1 information decaying faster than deep-book information.

3.3 Robustness Checks

Per-day consistency. On each test day independently, count-based FV wins on 63 to 75% of instruments at L5, $H = 50$. The advantage is present on both days.

	L1				L5			
H	ΔIC	t	p	Win%	ΔIC	t	p	Win%
10	+.016	+7.3	$< 10^{-4}$	77%	+.045	+18.8	$< 10^{-4}$	94%
20	+.019	+6.9	$< 10^{-4}$	70%	+.037	+12.3	$< 10^{-4}$	90%
50	+.015	+3.5	.0006	66%	+.027	+7.4	$< 10^{-4}$	74%
100	+.003	+0.6	.58	49%	+.018	+2.9	.004	59%

Table 3: Paired t-test of IC(count) − IC(quantity) across 185 instruments. Wilcoxon signed-rank test (not shown) agrees at every horizon.

Equities vs futures. For equities, the count advantage holds at both L1 and L5. For NIFTY and BANKNIFTY futures, count wins at L5 (IC +0.086 vs +0.066 at $H = 50$) but not at L1 (IC +0.005 vs +0.020). This is consistent with futures L1 being dominated by HFT market makers whose rapid order cycling inflates the count without reflecting genuine participation diversity. At deeper levels, where slower institutional flow sits, the count signal recovers.

The futures L1 result initially concerned me, as it seemed to contradict the hypothesis. But thinking about who populates L1 on NIFTY futures, HFT firms cycling orders at sub-millisecond speed, it makes sense that count at L1 reflects algorithmic activity rather than participation diversity. The signal reappearing at L5 is reassuring.

Spread conditioning. The count advantage at L5 is +0.022 IC for tight-spread stocks (<2 bps), +0.026 for medium (2 to 5 bps), and +0.043 for wide-spread stocks (>5 bps). The signal is strongest for illiquid stocks, where algorithmic fragmentation is less prevalent and order count more genuinely reflects independent decisions.

Fragmentation control. Splitting observations by average order size (above vs below median), the count-based FV achieves IC = +0.109 when average order size is small and +0.097 when large. The signal works in both regimes, ruling out pure order fragmentation as the explanation.

4. Order Count Flow

4.1 Queue Dynamics

We test a causal variant: does a *change* in order count over the last 20 ticks predict the forward return over the next 50 ticks? The order count flow (net change in bid minus ask order counts) achieves mean IC = +0.047 across 183 equities ($t = 9.3$, $p < 10^{-4}$). The equivalent quantity flow achieves IC = +0.023 ($t = 4.8$, $p < 10^{-4}$). Count flow is twice as predictive, with a significant difference of +0.024 ($t = 5.4$, $p < 10^{-4}$).

This is the test that convinced me the result is real. Changes in order count predicting future returns, and doing so more strongly than changes in quantity, suggests a good information channel.

4.2 Signed Count Flow vs Signed Volume Flow

Replacing signed volume (the standard OFI measure from Cont et al. [3]) with signed order count reveals a striking divergence (Figure 2):

Signed volume flow has negative IC (−0.015 to −0.021): when buying volume is high, the price tends to fall (mean reversion after large market orders). Signed count flow has positive IC (+0.014 to +0.032): when many separate trades are buyer-initiated, the price continues upward. It is not the size of the flow but the number of independent decisions that predicts direction.

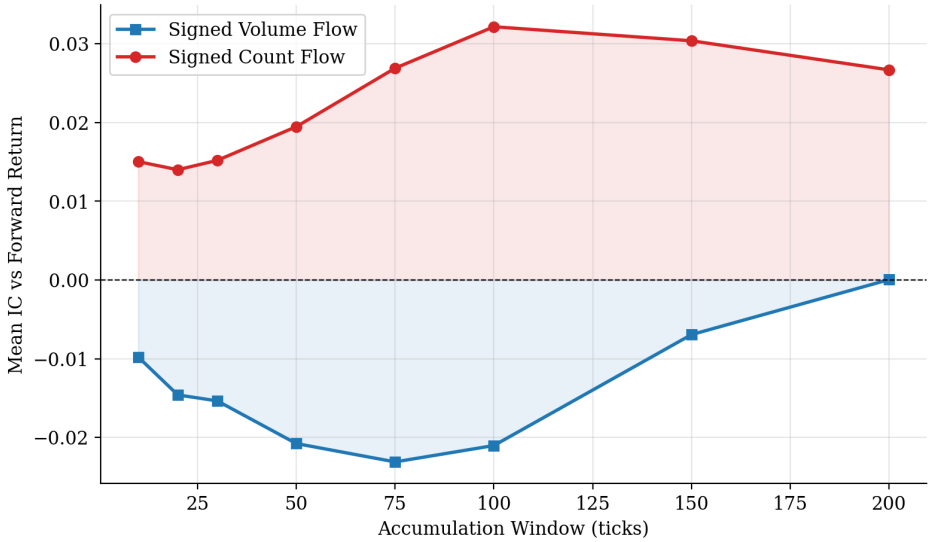

Figure 2: Signed count flow predicts returns in the correct direction. Signed volume flow predicts the wrong direction. The divergence widens with longer accumulation windows.

5. Supporting Analysis

5.1 Information Decay by Depth Level

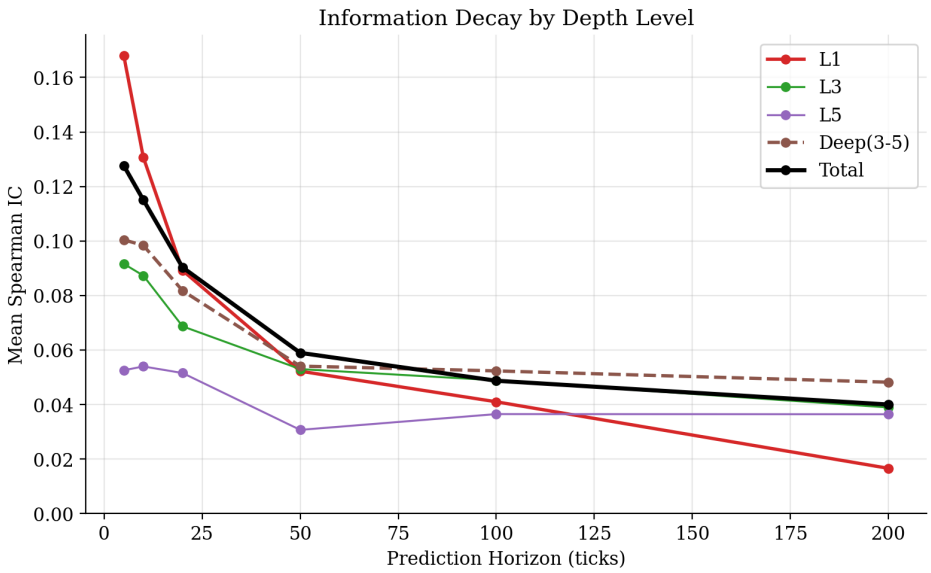

Figure 3: Spearman IC of OBI at each depth level vs forward return, averaged across 183 equities. L1 information decays with a half-life of approximately 4 ticks, while L5 persists with a half-life near 18 ticks.

L1 dominates at short horizons (IC $= 0.168$ at $H = 5$) but decays to 0.041 at $H = 100$. L3 starts weaker (0.087 at $H = 5$) but persists longer (0.049 at $H = 100$), crossing L1 around $H \approx 100$. This is consistent with the finding that the count advantage is strongest at L5 and at longer horizons: deeper levels carry slower but more persistent information from less transient participants.

I initially hypothesised that deeper book levels would dominate L1 for return prediction. The level-by-level analysis revealed the following: each level has a characteristic timescale, and the optimal weighting depends on your trading horizon. This led me to test the multi-level count-weighted microprice, which outperforms all single-level variants.

5.2 Cross-Sectional Signal

At each evaluation point, ranking all 183 equities by their z-scored order count imbalance (OCI) and forming a quintile long/short portfolio yields market-neutral returns of +1.48 bps at 30 seconds (86.6% hit rate) increasing to +2.13 bps at 5 minutes (68.8% hit rate). OCI-ranked portfolios are approximately 50% stronger than quantity-ranked portfolios at every horizon (Figure 4).

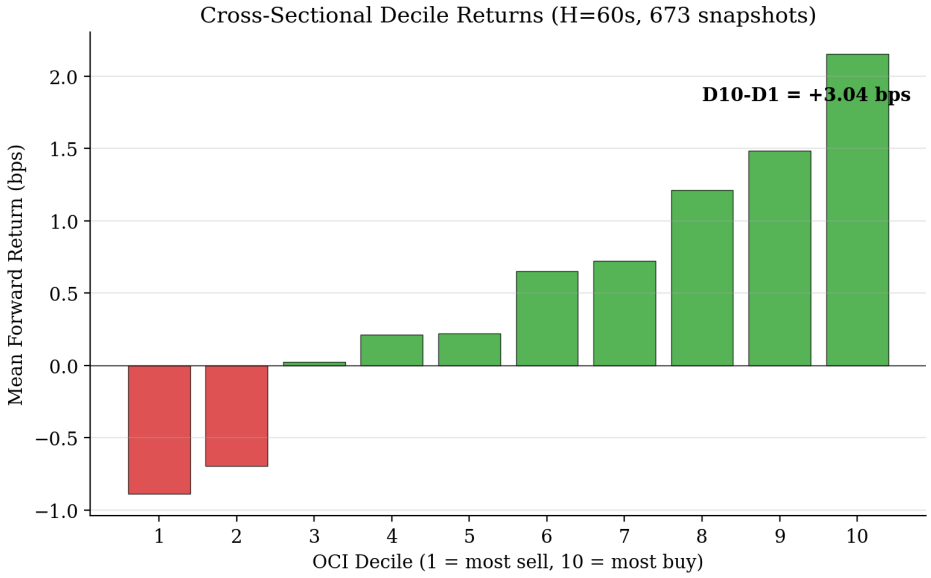

Figure 4: Cross-sectional decile returns: stocks ranked by order count imbalance at each snapshot, forward returns measured over 60 seconds. 673 non-overlapping snapshots across test days.

5.3 Machine Learning Enhancement

To test whether the count-based signal combines with temporal features, we train a 2-layer GRU (64 hidden units) on 20-tick sequences across 183 equities. The feature set includes 17 standard LOB features plus 6 order flow features including order count imbalance (Figure 5).

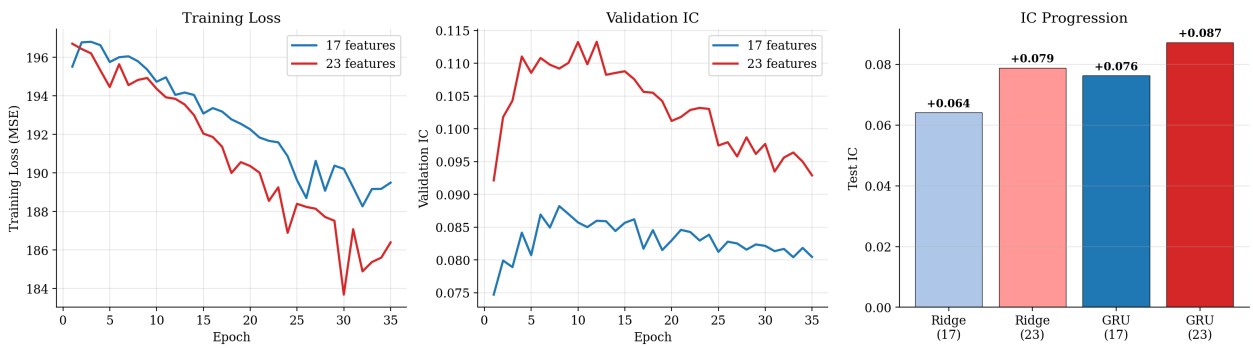

Figure 5: GRU training diagnostics and IC progression. Order flow features add +23% to Ridge. Temporal modelling adds +19%. Combined: +36% over baseline.

6. Proposed Research

These findings are preliminary, based on 5 days of data from a single market. The proposed internship would extend and validate this work on Teza's infrastructure.

Month 1: Cross-Market Validation. Test whether the count-based advantage holds on US equities, futures, and FX. US markets have 16+ venues, so order count on any single venue understates true participation. With consolidated L3 data and order IDs, the analysis can be refined to measure unique participant count, which should be an even stronger signal than raw order count.

Model	Features	Test IC	D10-D1 (bps)
Ridge	17 (standard LOB)	+0.064	
Ridge	23 (+order flow)	+0.079	
GRU	17 (standard LOB)	+0.076	+1.50
GRU	23 (+order flow)	**+0.087**	+1.64

Table 4: Return prediction at $H = 50$. Walk-forward validation: train on days 1-2, validate on day 3, test on days 4-5. Order count imbalance (IC = +0.094 as a single feature) is the strongest predictor in the expanded set, exceeding quantity-based OBI (+0.052) by 80%.

Month 2: Production Fair Value Model. Integrate count-based weighting into a state-conditional GRU fair value model. In different regimes (high volatility, wide spreads, post-event), the optimal blend of count and quantity may differ. The architecture learns these weights from data. Benchmark against Teza's existing fair value model on historical data.

Month 3: Execution and Backtesting. Designing the live-test (or paper trading) system for low latency from scratch. ONNX export for production-latency inference. Backtest with realistic market impact and queue position modelling. Measure the P&L impact of switching from quantity-based to count-based fair value in the quoting engine.

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